Aditi Sarker

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https://github.com/krishnaaditi Google_Scholar_Aditi

Research Interests

My research focuses on Multimodal Large Language Models (MLLMs), particularly in hallucination mitigation, modality alignment, and scalable learning. Currently, I am developing a Hierarchical Fine-Tuning approach to enhance modality fusion and reduce errors. My future work includes Direct Preference Optimization (DPO) for improved optimization and adversarial attack analysis to strengthen MLLM security and robustness.

Education

Wayne State University — **Aug. 2023 – Present(3.97 out of 4.00)**
PhD in Computer Science — *Detroit, MI, USA*

Ph.D. Candidate (Expected 2028)

- Qualifying Exam (Survey Report) Title: The Role of Contrastive Learning in Multimodal Large Language Models (MLLMs).[Report]

Jahangirnagar University — **Apr. 2017 – Nov. 2018(4.00 out of 4.00)**
MSc. in Computer Science and Engineering — *Dhaka, Bangladesh*

- Thesis Title: Simulation of hazy image and validation of haze removal technique.

Jahangirnagar University — **Dec. 2011 – Dec.2016(3.78 out of 4.00)**
BSc.in Computer Science and Engineering — *Dhaka, Bangladesh*

Work Experiences

Graduate Teaching Assistant — **August 2026 – Present**
Wayne State University — *Detroit, Michigan*

- Developed an algorithm on Hybrid Federated Learning and experiment.
- Developed algorithm on Federated Compositional Optimization and experiment.

Graduate Teaching Assistant — **May 2026 – August 2026**
Wayne State University — *Detroit, Michigan*

- Developed an algorithm on Hybrid Federated Learning and experiment.
- Developed algorithm on Federated Compositional Optimization and experiment.

Graduate Teaching Assistant — **August 2025 – May 2026**
Wayne State University — *Detroit, Michigan*

- Developed an algorithm on Hybrid Federated Learning and experiment.
- Developed algorithm on Federated Compositional Optimization and experiment.

Graduate Research Assistant — **May 2025 – August 2025**
Wayne State University — *Detroit, Michigan*

- Developed an algorithm on Hybrid Federated Learning and experiment.
- Developed algorithm on Federated Compositional Optimization and experiment.

Graduate Research Assistant — **May 2024 – August 2024**
Wayne State University — *Detroit, Michigan*

- Developed an algorithm on Hybrid Federated Learning and experiment.
- Developed algorithm on Federated Compositional Optimization and experiment.

Graduate Teaching Assistant — **August 2023 – May 2024**
Wayne State University — *Detroit, Michigan*

- Course: Problem Solving and Programming.
- Took lab class, tough programming in c++ language, grading, consultation, exam taking.

Assistant Professor — **December 2022 – August 2023**
Comilla University — *Cumilla Bangladesh*

- Course: Data Mining, Artificial Intelligences, Algorithm Analysis and Design, Theory of Computation , Software Engineering.
- Took classes, Took lab classes, grading, consultation, exam taking.

Lecturer **December 2019 – December 2022**
Comilla University *Cumilla, Bangladesh*

- Course: Graph Theory, Object Oriented Programming, Deep Learning.
- Took classes, Took lab classes, grading, consultation, exam taking.

Fellowships and Awards

Thomas C. Rumble Fellowship **August 2024 – Present**
Wayne State University

WiCV @ CVPR 2026 Travel Grant **June 2026**
Women in Computer Vision (WiCV), CVPR 2026 *Denver, Colorado, USA*

Ongoing Research

LIFE: Local Intrinsic-Dimension for Finding and Repairing Sample-Specific Hallucination-Critical Representations i

- Developing an automatic framework to identify sample-specific hallucination-critical representations in vision-language models using local intrinsic dimension and repair them to mitigate hallucinations while preserving model performance.

Publications

Manuscripts Under Review

- **Sarker, A.**, Sultan, R. I., Zhu, H., Zhu, D., & Khanduri, P. *Hallucination Mitigation for Large Vision Language Models via Implicit Feature Stabilization.* Under review at **AAAI 2027**.
- **Sarker, A.**, Shah, N., Sultan, R. I., Jang, R., Zhu, D., & Khanduri, P. *Partition-Aware Unlearning for Removing Spurious Correlations in Large Vision-Language Models.* Under review at NeurIPS 2026.

Journal Articles

- Khanduri, P., Li, C., Sultan, R. I., **Sarker, A.**, Qiang, Y., Kliewer, J., & Zhu, D. *On Federated Compositional Optimization: Algorithms, Analysis, and Guarantees.* **Transactions on Machine Learning Research (TMLR)**, 2026.
- **Sarker, A.**, Chakraborty, P., Sha, S. S., Khatun, M., Hasan, M. R., & Banerjee, K. (2020). Improvised technique for analyzing data and detecting terrorist attack using machine learning approach based on twitter data. Journal of Computer and Communications, 8 (7),50–62. doi:10.4236/jcc.2020.87
- **Sarker, A.**, Akter, M., & Uddin, M. S. (2019). Simulation of hazy image and validation of haze removal technique. Journal of Computer and Communications, 7 (2), 62–72. doi:10.4236/jcc.2019.72005

Conference Proceedings

- Chandra Das, S., **Sarker, A.**, Saha, S., & Chakraborty, P. (2022). Covid-19 in bangladesh: An exploratory data analysis and prediction of neurological syndrome using machine learning algorithms based on comorbidity. In V. Skala, T. P. Singh, T. Choudhury,R. Tomar, M. Abul Bashar (Eds.), Machine intelligence and data science applications (pp. 595–608).doi:10.1007/978-981-19-2347-0
- Uddin, M. S., Gautam, B., **Sarker, A.**, Akter, M., & Haque, M. R. (2017). Image-based automated haze removal using dark channel prior. In 2017 ieee region 10 humanitarian technology conference (r10-htc) (pp. 412–415).doi:10.1109/R10-HTC.2017.828
- Feroz, M. A., Sultana, M., Hasan, M. R., **Sarker, A.**, Chakraborty, P., & Choudhury, T.(2022). Object detection and classification from a real-time video using ssd and yolo models. In A. K. Das, J. Nayak, B. Naik, S. Dutta, D. Pelusi (Eds.), Computational intelligence in pattern recognition (pp. 37–47). Singapore: Springer Singapore.

- Chakraborty, P., Roy, S., Sumaiya, S. N., & **Sarker, A.** (2023). Handwritten character recognition from image using cnn. In Micro-electronics and telecommunication engineering: proceedings of 6th ICMETE 2022 (pp. 37-47). Singapore: Springer Nature Singapore.
- Chakraborty, P., Yousuf, M. A., Islam, S., Khatun, M., **Sarker, A.**, & Rahman, S. (2022). Analysis of Texture Feature Extraction Technique in Image Processing. In Distributed Sensing and Intelligent Systems: Proceedings of ICDSIS 2020 (pp. 651-665). Cham: Springer International Publishing.

Books and Chapters

- Uddin, M. S., Shorif, S. B., & **Sarker, A.** (2021). Role-framework of artificial intelligence in combating the covid-19 pandemic (M. A. R. Ahad A. Inoue, Eds.). doi:10.1007/978-3-030-75490-7

Extended Abstract

- **Sarkar, A.**, Rahman, M., & Uddin, M. S. (2020). Automated detection of plan diseases using r- cnn algorithms. IEEE CSBC Winter Symposium.
- Sheikh, R., **Sarkar, A.**, Uddin, M. S. (2020). Simulation of haze on real world images with generative adversarial network. IEEE CSBC Winter Symposium.

Projects(PhD)

Attention-Enhanced GraphSAGE with Skip-Fusion | *Python, C++* **January-April 2025**

- This project develops an attention-enhanced GraphSAGE model with skip-fusion that adaptively weights sampled neighbors and preserves node-specific features, improving robustness and representation quality in inductive graph learning while maintaining scalability.

What Can Transformers Learn In-Context? | *Python, C++* **January-April 2024**

- The project demonstrate that Transformers can be trained to master more intricate function classes, such as sparse linear functions, decision trees, and two-layer neural networks, achieving performance levels that match or surpass those of task-specific learning algorithms.

CARLA Self-Driving Motion Planning in Autonomous Driving | *Python, C++* **January-April 2024**

- We employ a functional motion planning stack. This is accomplished using behavioral planning logic, path selection, velocity profile development, and static collision checking. This project develops a functional motion planning stack for self-driving cars that allows them to avoid both static and dynamic barriers while tracking the center line of a lane and handling stop signs.

Technical Skills

Languages: Python, Java, C, HTML/CSS, JavaScript, SQL
Developer Tools: VS Code, Eclipse, Google Cloud Platform, Android Studio
Technologies/Frameworks: Linux, Jenkins, GitHub, JUnit, WordPress

Certified Courses

Advance Machine Learning and Data Science **2020**
AILABS, HR Venture *Duration: 6 month*

Leadership / Extracurricular

CSE Society **August 2020 – August 2023**
Member *Comilla University*

Exam Committee **August 2020 – August 2023**
Member *Comilla University*

IWCI Workshop **2016**
Organiser *Jahangirnagar University*

References

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Wayne State University, MI
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